International Baccalaureate®
Baccalauréat International
Bachillerato Internacional

**CHEMISTRY**
**STANDARD LEVEL**
**PAPER 1**

SPECIMEN PAPER

45 minutes

---

INSTRUCTIONS TO CANDIDATES

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- The periodic table is provided for reference on page 2 of this examination paper.
- The maximum mark for this examination paper is *[30 marks]*.


© International Baccalaureate Organization 2014

# The Periodic Table

Atomic number

**Element**

Relative atomic mass

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1<br>**H**<br>1.01 | | | | | | | | | | | | | | | | | 2<br>**He**<br>4.00 |
| 2 | 3<br>**Li**<br>6.94 | 4<br>**Be**<br>9.01 | | | | | | | | | | | 5<br>**B**<br>10.81 | 6<br>**C**<br>12.01 | 7<br>**N**<br>14.01 | 8<br>**O**<br>16.00 | 9<br>**F**<br>19.00 | 10<br>**Ne**<br>20.18 |
| 3 | 11<br>**Na**<br>22.99 | 12<br>**Mg**<br>24.31 | | | | | | | | | | | 13<br>**Al**<br>26.98 | 14<br>**Si**<br>28.09 | 15<br>**P**<br>30.97 | 16<br>**S**<br>32.07 | 17<br>**Cl**<br>35.45 | 18<br>**Ar**<br>39.95 |
| 4 | 19<br>**K**<br>39.10 | 20<br>**Ca**<br>40.08 | 21<br>**Sc**<br>44.96 | 22<br>**Ti**<br>47.87 | 23<br>**V**<br>50.94 | 24<br>**Cr**<br>52.00 | 25<br>**Mn**<br>54.94 | 26<br>**Fe**<br>55.85 | 27<br>**Co**<br>58.93 | 28<br>**Ni**<br>58.69 | 29<br>**Cu**<br>63.55 | 30<br>**Zn**<br>65.38 | 31<br>**Ga**<br>69.72 | 32<br>**Ge**<br>72.63 | 33<br>**As**<br>74.92 | 34<br>**Se**<br>78.96 | 35<br>**Br**<br>79.90 | 36<br>**Kr**<br>83.90 |
| 5 | 37<br>**Rb**<br>85.47 | 38<br>**Sr**<br>87.62 | 39<br>**Y**<br>88.91 | 40<br>**Zr**<br>91.22 | 41<br>**Nb**<br>92.91 | 42<br>**Mo**<br>95.96 | 43<br>**Tc**<br>(98) | 44<br>**Ru**<br>101.07 | 45<br>**Rh**<br>102.91 | 46<br>**Pd**<br>106.42 | 47<br>**Ag**<br>107.87 | 48<br>**Cd**<br>112.41 | 49<br>**In**<br>114.82 | 50<br>**Sn**<br>118.71 | 51<br>**Sb**<br>121.76 | 52<br>**Te**<br>127.60 | 53<br>**I**<br>126.90 | 54<br>**Xe**<br>131.29 |
| 6 | 55<br>**Cs**<br>132.91 | 56<br>**Ba**<br>137.33 | 57 †<br>**La**<br>138.91 | 72<br>**Hf**<br>178.49 | 73<br>**Ta**<br>180.95 | 74<br>**W**<br>183.84 | 75<br>**Re**<br>186.21 | 76<br>**Os**<br>190.23 | 77<br>**Ir**<br>192.22 | 78<br>**Pt**<br>195.08 | 79<br>**Au**<br>196.97 | 80<br>**Hg**<br>200.59 | 81<br>**Tl**<br>204.38 | 82<br>**Pb**<br>207.2 | 83<br>**Bi**<br>208.98 | 84<br>**Po**<br>(209) | 85<br>**At**<br>(210) | 86<br>**Rn**<br>(222) |
| 7 | 87<br>**Fr**<br>(223) | 88<br>**Ra**<br>(226) | 89 ‡<br>**Ac**<br>(227) | 104<br>**Rf**<br>(267) | 105<br>**Db**<br>(268) | 106<br>**Sg**<br>(269) | 107<br>**Bh**<br>(270) | 108<br>**Hs**<br>(269) | 109<br>**Mt**<br>(278) | 110<br>**Ds**<br>(281) | 111<br>**Rg**<br>(281) | 112<br>**Cn**<br>(285) | 113<br>**Unt**<br>(286) | 114<br>**Uug**<br>(289) | 115<br>**Uup**<br>(288) | 116<br>**Uuh**<br>(293) | 117<br>**Uus**<br>(294) | 118<br>**Uuo**<br>(294) |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| † | 58<br>**Ce**<br>140.12 | 59<br>**Pr**<br>140.91 | 60<br>**Nd**<br>144.24 | 61<br>**Pm**<br>(145) | 62<br>**Sm**<br>150.36 | 63<br>**Eu**<br>151.96 | 64<br>**Gd**<br>157.25 | 65<br>**Tb**<br>158.93 | 66<br>**Dy**<br>162.50 | 67<br>**Ho**<br>164.93 | 68<br>**Er**<br>167.26 | 69<br>**Tm**<br>168.93 | 70<br>**Yb**<br>173.05 | 71<br>**Lu**<br>174.97 |
| ‡ | 90<br>**Th**<br>232.04 | 91<br>**Pa**<br>231.04 | 92<br>**U**<br>238.03 | 93<br>**Np**<br>(237) | 94<br>**Pu**<br>(244) | 95<br>**Am**<br>(243) | 96<br>**Cm**<br>(247) | 97<br>**Bk**<br>(247) | 98<br>**Cf**<br>(251) | 99<br>**Es**<br>(252) | 100<br>**Fm**<br>(257) | 101<br>**Md**<br>(258) | 102<br>**No**<br>(259) | 103<br>**Lr**<br>(262) |

**1.** What is the total number of atoms in 0.50 mol of 1,4-diaminobenzene, $H_2NC_6H_4NH_2$? (Avogadro's constant (L or $N_A$) = $6.0 \times 10^{23}$ $mol^{-1}$.)

A. $16.0 \times 10^{23}$

B. $48.0 \times 10^{23}$

C. $96.0 \times 10^{23}$

D. $192.0 \times 10^{23}$

**2.** What is the sum of the coefficients when the equation for the combustion of ammonia is balanced using the smallest possible whole numbers?

$$___\ NH_3(g) + ___\ O_2(g) \rightarrow ___\ N_2(g) + ___\ H_2O(g)$$

A. 6

B. 12

C. 14

D. 15

**3.** Which changes of state are endothermic processes?

I. Condensing

II. Melting

III. Subliming

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

**4.** 5.00 g of calcium carbonate, when heated, produced 2.40 g of calcium oxide. Which is the correct expression for the percentage yield of calcium oxide? ($M_r(CaCO_3) = 100$; $M_r(CaO) = 56$.)

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

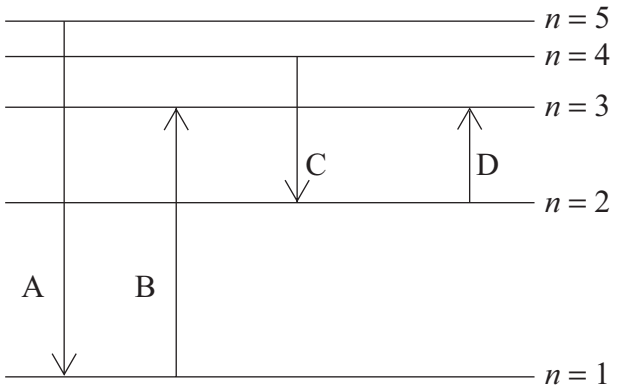
A. $\dfrac{56 \times 5.00 \times 100}{2.40}$

B. $\dfrac{2.40 \times 100 \times 100}{56 \times 5.00}$

C. $\dfrac{56 \times 5.00 \times 100}{2.40 \times 100}$

D. $\dfrac{2.40 \times 100}{56 \times 5.00}$

**5.** Which electronic transition would absorb the radiation of the shortest wavelength?




**6.** What are the numbers of protons, neutrons and electrons in the ion $^{238}_{92}X^{2+}$?

| | Protons | Neutrons | Electrons |
|---|---|---|---|
| A. | 146 | 92 | 144 |
| B. | 92 | 146 | 90 |
| C. | 92 | 146 | 94 |
| D. | 92 | 238 | 90 |

**7.** Which element is in the f-block of the periodic table?

A. Be

B. Ce

C. Ge

D. Re

**8.** Which property increases down group 1 of the periodic table?

A. Melting point

B. First ionization energy

C. Atomic radius

D. Electronegativity

**9.** Which is the best description of ionic bonding?

A. Electrostatic attraction between oppositely charged ions

B. Electrostatic attraction between positive ions and electrons

C. Electrostatic attraction of nuclei towards shared electrons in the bond between the nuclei

D. Electrostatic attraction between nuclei

**10.** Which has bonds with the greatest covalent character?

A. $SrCl_2$

B. $SiCl_4$

C. $SnCl_2$

D. Sn

**11.** Which bond is the **least** polar?

A. C=O in $CO_2$

B. C–H in $CH_4$

C. C–Cl in $CCl_4$

D. N–H in $CH_3NH_2$

**12.** Which substance has a high melting point and does not conduct electricity in any state?

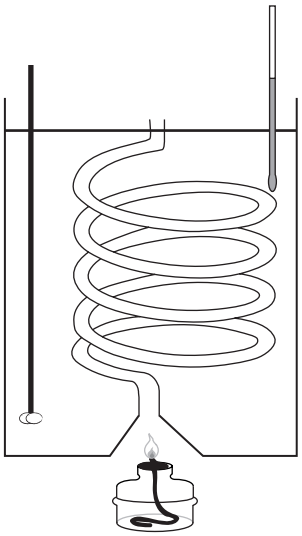
A. $PbBr_2$

B. Fe

C. NaCl

D. $SiO_2$

**13.** When 0.46 g of ethanol is burned under a water-filled calorimeter, the temperature of 500 g of water is raised by 3.0 K. (Molar mass of ethanol = 46 g mol$^{-1}$; specific heat capacity of water = 4.18 J g$^{-1}$ K$^{-1}$; $q = mc\Delta T$.)



What is the expression for the enthalpy of combustion, $\Delta H_c$, in kJ mol$^{-1}$?

A. $-\dfrac{500 \times 4.18 \times 3.0 \times 46}{0.46}$

B. $-\dfrac{500 \times 4.18 \times (273 + 3.0) \times 46}{0.46 \times 1000}$

C. $-\dfrac{500 \times 4.18 \times 3.0 \times 46}{0.46 \times 1000}$

D. $-\dfrac{0.46 \times 1000}{500 \times 4.18 \times 3.0 \times 46}$

**14.** Which reaction represents the average bond enthalpy of the C–H bond?

A. $\frac{1}{4}CH_4(g) \rightarrow \frac{1}{4}C(g) + \frac{1}{2}H_2(g)$

B. $\frac{1}{4}CH_4(g) \rightarrow \frac{1}{4}CH_2(g) + \frac{1}{4}H_2(g)$

C. $\frac{1}{4}CH_4(g) \rightarrow \frac{1}{4}C(g) + H(g)$

D. $\frac{1}{4}CH_4(g) \rightarrow \frac{1}{4}C(s) + H(g)$

**15.** Given the following information, what is the standard enthalpy of formation, $\Delta H^{\ominus}_{f}$, of methane?

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = \text{E kJ}$$
$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = \text{F kJ}$$
$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H = \text{G kJ}$$

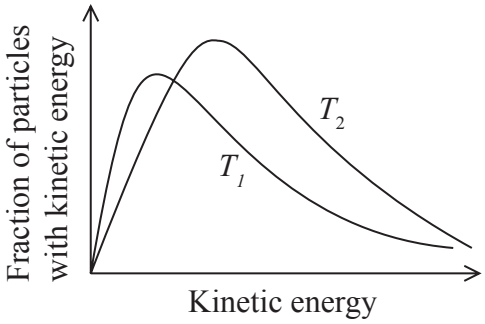
A. E + F + G

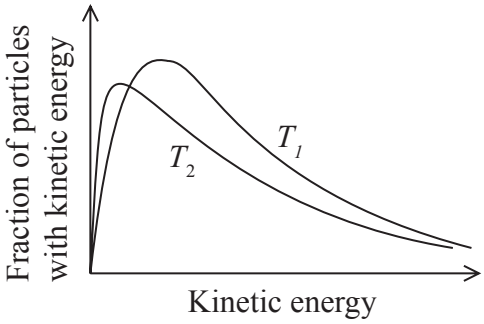
B. E + F – G

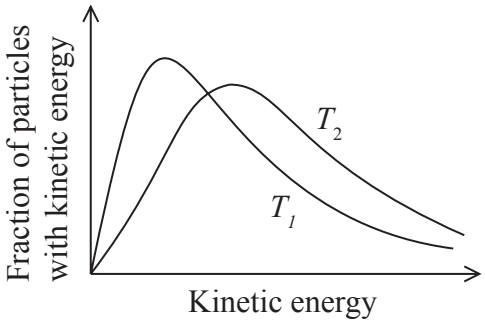
C. E + 2F + G

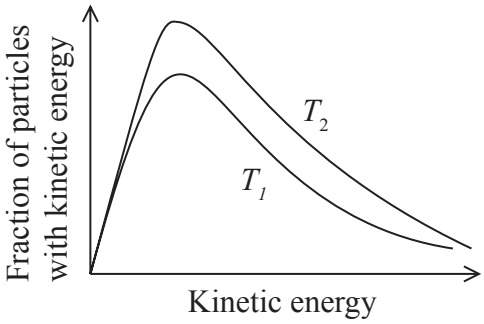
D. E + 2F – G

**16.** Which graph shows the Maxwell-Boltzmann energy distribution of a same amount of a gas at two temperatures, where $T_2$ is greater than $T_1$?

A.



B.



C.



D.


**17.** Which changes increase the rate of this reaction, other conditions remaining constant?

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

I. Using larger lumps of calcium carbonate

II. Increasing the temperature of the reaction mixture

III. Increasing the concentration of hydrochloric acid

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

**18.** Which conditions give the greatest equilibrium yield of methanal, $H_2CO(g)$?

$$CO(g) + H_2(g) \rightleftharpoons H_2CO(g) \qquad \Delta H = -1.8\,kJ$$

| | Pressure | Temperature |
|---|---|---|
| A. | high | low |
| B. | high | high |
| C. | low | high |
| D. | low | low |

**19.** Which of the following is **not** amphiprotic?

A. $H_2O$

B. $HPO_4^{2-}$

C. $H_2PO_4^-$

D. $H_3O^+$

**20.** Which compound will react with dilute hydrochloric acid, HCl(aq), to give off a gas?

A. $Cu_2O(s)$

B. $Cu(OH)_2(s)$

C. $CuCO_3(s)$

D. $CuO(s)$

**21.** The equations below represent reactions involved in the Winkler method for determining the concentration of dissolved oxygen in water:

$$2Mn(OH)_2(s) + O_2(aq) \rightarrow 2MnO(OH)_2(s)$$
$$MnO(OH)_2(s) + 2H_2SO_4(aq) \rightarrow Mn(SO_4)_2(s) + 3H_2O(l)$$
$$Mn(SO_4)_2(s) + 2I^-(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2SO_4^{2-}(aq)$$
$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

What is the amount, in mol, of thiosulfate ions, $S_2O_3^{2-}(aq)$, needed to react with the iodine, $I_2(aq)$, formed by 1.00 mol of dissolved oxygen?

A. 2.00

B. 3.00

C. 4.00

D. 6.00

**22.** What are the products when molten sodium chloride is electrolysed?

| | **Cathode** | **Anode** |
|---|---|---|
| A. | hydrogen | chlorine |
| B. | sodium | chloride |
| C. | sodium | chlorine |
| D. | chlorine | sodium |

**23.** Which is propyl propanoate?

A. $CH_3CH_2CH_2OOCCH_2CH_3$

B. $CH_3CH_2CH_2COOCH_2CH_3$

C. $CH_3CH_2CH_2COCH_2CH_3$

D. $CH_3CH_2CH_2OCH_2CH_2CH_3$

**24.** Which are consecutive members of a homologous series?

A. $CH_4$, $CH_3Cl$, $CH_2Cl_2$

B. HCOOH, $CH_3COOH$, $C_2H_5COOH$

C. $C_2H_2$, $C_2H_4$, $C_2H_6$

D. HCOOH, HCHO, $CH_3OH$

**25.** Which could form an addition polymer?

A. $H_2NCH_2CHCHCH_2NH_2$

B. $H_2N(CH_2)_6CO_2H$

C. $HO(CH_2)_2CO_2H$

D. $H_2N(CH_2)_6NH_2$

**26.** A compound decolorizes bromine water in the dark. Which statement is correct?

A. It contains C=C and is an alkene.

B. It contains C–C and is an alkene.

C. It contains C=C and is an alkane.

D. It contains C–C and is an alkane.

**27.** How can a systematic error be minimized?

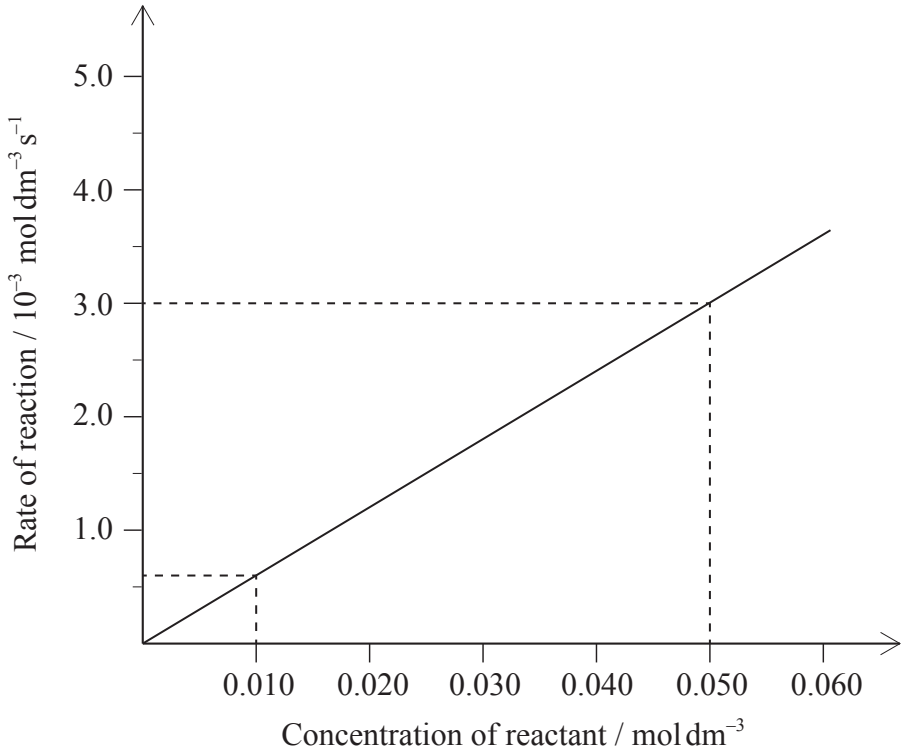
A. By taking the reading many times

B. By repeating the experiment many times

C. By using a more accurate measuring device

D. By evaluating and modifying the method

**28.** Which combination in the table correctly states the value and units of the gradient?




| | Value | Units |
|---|---|---|
| A. | $\frac{3.0\times10^{-3}-0.6\times10^{-3}}{0.050-0.010}$ | $s^{-1}$ |
| B. | $\frac{3.0\times10^{-3}-0.6\times10^{-3}}{0.050-0.010}$ | s |
| C. | $\frac{0.050-0.010}{3.0\times10^{-3}-0.6\times10^{-3}}$ | $s^{-1}$ |
| D. | $\frac{0.050-0.010}{3.0\times10^{-3}-0.6\times10^{-3}}$ | s |

**29.** Which part of the electromagnetic spectrum is used by $^{1}H$ NMR spectroscopy?

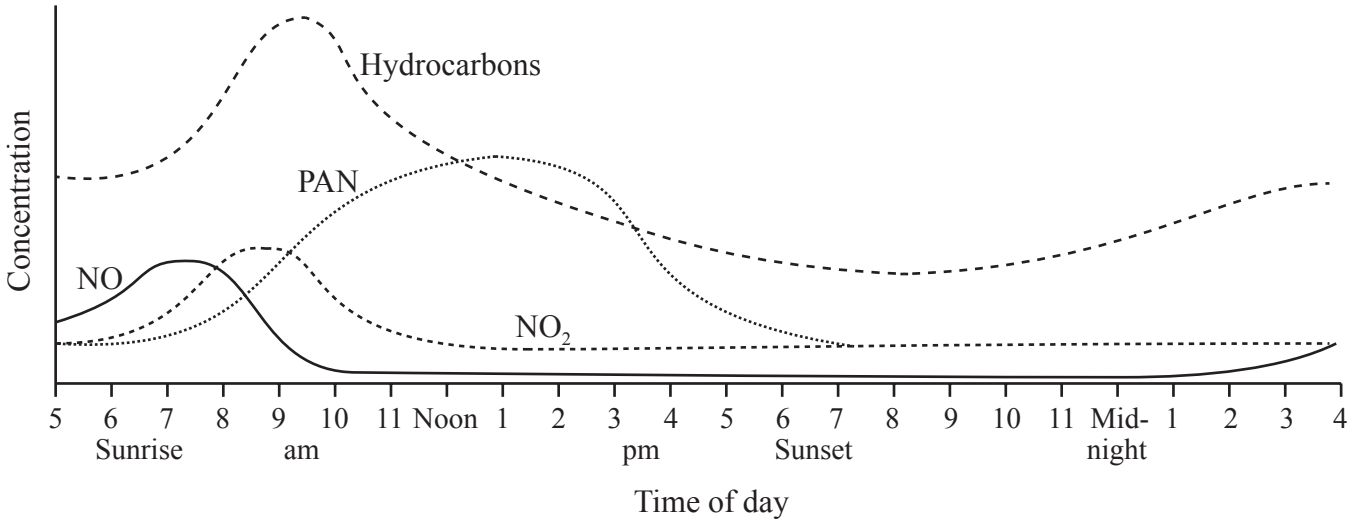
A. γ rays

B. X-rays

C. Microwaves

D. Radio waves

**30.** The graph shows the concentration of some pollutants in a city over a 24-hour period.




Which of the following could **not** be inferred from the graph?

A. Hydrocarbons cause less harm to health than PAN.

B. An increase in hydrocarbons is caused by the morning rush hour.

C. PAN concentration increases as the intensity of sunlight increases.

D. $NO_2$ production follows the production of NO.